

Department of Mechanical and Production Engineering

Aarhus University

Fremstillingsprocesser

Author	Noah Rahbek Bigum Hansen	202405538
Course	Fremstillingsprocesser	290221U001
Instructor	Jinghao Li & Tito Andriollo	
Semester	Semester 5	
Date	August 27, 2026	



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1 Introduction

1.1 What Is Manufacturing

As a technological process, manufacturing is the application of a physical and/or chemical process to alter the geometry, properties, and/or appearance of a starting material to make parts or products. Economically, a manufacturing process can be seen as the transformation of materials into items of greater value by one or more processing and/or assembly operations.

1.2 Production Quantity and Product Variety

The *Production Quantity* Q is the quantity of products made by a factory. Annual quantities are classified into three ranges:

- Low Production — 1–100 units
- Medium Production — 100–10 000 units
- High/Mass Production — 10 000+ units

The *Product Variety* P refers to different product types or models produced in the plant. These different products often have different features, are intended for different markets, or have different parts. The number of different product types made each year in a factory can be counted yielding the product variety of that year.

Although product variety P is quantitative, it is much less exact than the production quantity Q because details on how much the designs differ is not captured accurately simply by the number of different designs. If the difference between products is small it is termed *Soft Product Variety* (e.g. storage options on a PC) and if the products differ substantially it is termed *Hard Product Variety* (e.g. a car company producing both semi-trucks and small cars).

1.3 Manufacturing Capability and Processes

Manufacturing capability is a measure of how “much” manufacturing is possible. This depends on both technological process capability, physical product limitations, or limited production capacity.

Material parameters are of high importance here. Most materials can be classified as either metals, ceramics, composites, or polymers. Each have different chemistry meaning their mechanical and physical properties are different.

Different manufacturing processes can be classified as shown on figure 1.1: [Classification of manufacturing processes](#) (Thorat 2026). The main focuses of this course will be on the shaping processes and permanent joining.

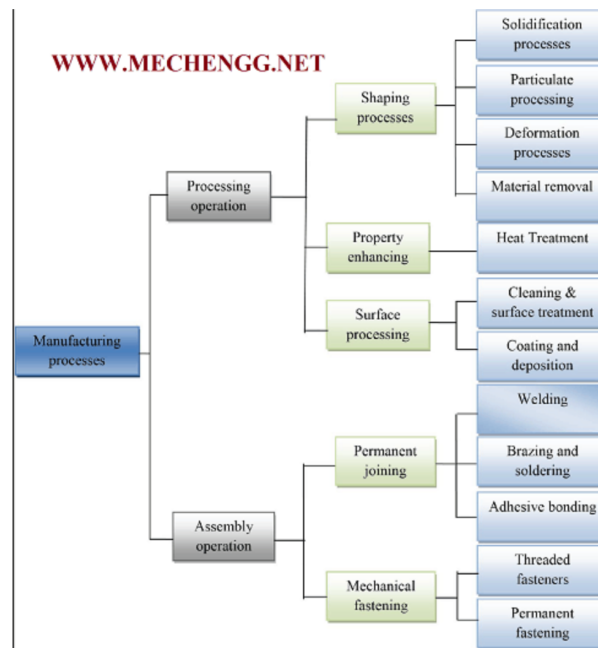


Figure 1.1: Classification of manufacturing processes (Thorat 2026)

1.4 Manufacturing Production Systems

Product development normally follows a so-called “S”-curve, where the performance starts of at a low point before growing quickly until the product has matured or the design has been optimized sufficiently that the increase in performance will wane off. This looks a bit like the sigmoid function, if performance is plotted with respect to time invested.

A production system may follow the model shown on figure 1.2: Model of a production system. Many factors here are able to be changed, e.g.

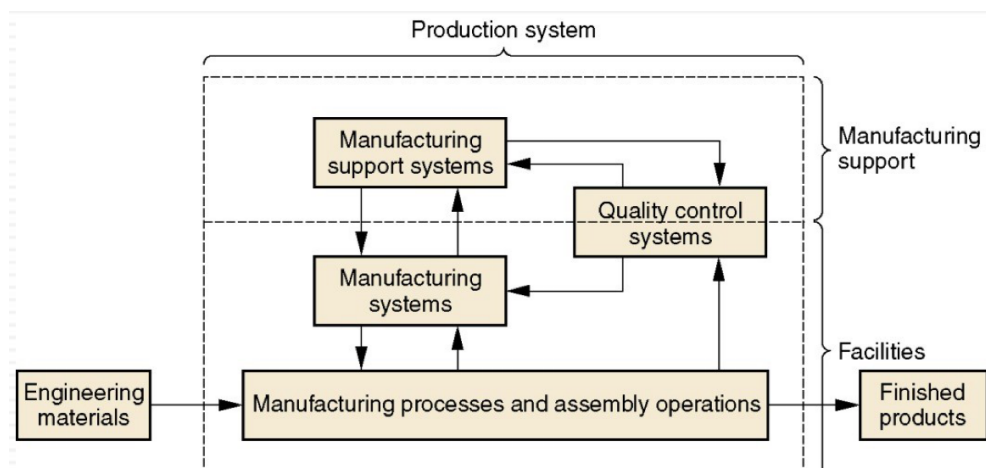


Figure 1.2: Model of a production system

- Product design
- Materials
- Labor costs
- Equipment

- Manufacturing costs

1.4.1 Low Production

Low Production systems are sometimes called *Job shops*. These job shops makes low quantities of specialized and highly-customized products, e.g. space capsules, prototype aircraft, special machinery, etc. The equipment at these places is general purpose but the labor force is highly skilled meaning job shops offer maximal flexibility.

1.4.2 Medium Production

Two different types of facility exists, depending on product variety. For medium to hard product variety, *batch production* is often used, where different setups are required between batches. For soft variety plants, *Cellular Manufacturing* is widespread. Here, worker cells are organized to process parts without setups between different part styles.

1.4.3 High Production

Also called *mass production*, this type of production is needed when the demand for product is high and the entire manufacturing system is dedicated to the production of that product. This also exists in two categories. The first is *Quantity Production*, i.e. mass production of single parts on a single or small number of machines, and the second is *Flow Line Production*, i.e. multiple machines or workstations arranged in sequence, as in a production line.

References

Thorat, Sachin (2026). *Introduction to Manufacturing Processes and Their Classification*. Learn-Mech.com. URL: <https://learnmech.com/types-manufacturing-processes/> (visited on 08/27/2026).